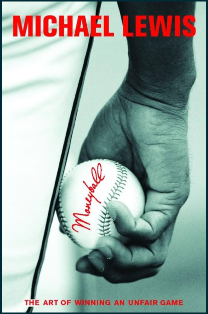
Winning baseball games by solving statistics puzzles

Scott Powers

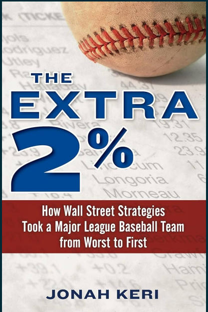
JHU ECE Seminar
September 18, 2025

RICE UNIVERSITY
Sport Analytics

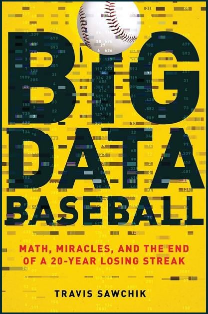
Books about Baseball Teams Using Analytics to Win



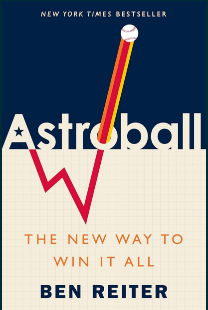
2002
Oakland
Athletics



2008
Tampa Bay
Rays



2013
Pittsburgh
Pirates

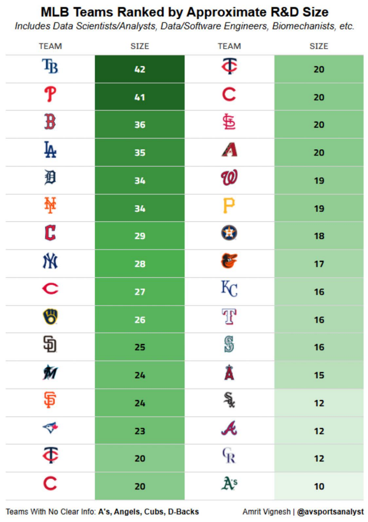


2017
Houston
Astros

MLB R&D Staff Sizes (according to Reddit)

Team	Analysts	Team	Analysts
American League		**National League**	
Yankees	20	Dodgers	20
Astros	15	Braves	15
Rays	15	Brewers	11
Angels	10	Reds	11
Tigers	9	Phillies	10
Rangers	8	Nationals	8
Mariners	7	Pirates	8
Royals	7	Padres	7
Twins	7	Cardinals	6
Blue Jays	6	Cubs	6
Red Sox	6	Giants	6
Indians	5	Marlins	6
Orioles	5	Dbacks	5
Athletics	3	Rockies	4
White Sox	2	Mets	3

2018

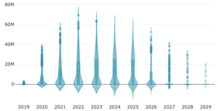


2025

What are they all doing?

Player Evaluation

Executives use modeling results to decide which players to target in trades and in free agency.

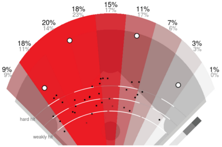


Player Development

Players use data-driven insights to tweak their game and achieve greater levels of performance.

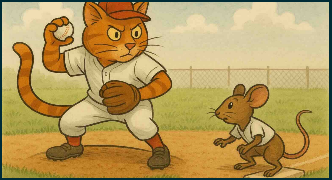
In-Game Strategy

Coaches use analytics tools to position fielders and decide when to make substitutions.

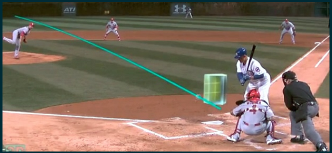


Three Puzzles

I: Runner-pitcher strategy under pickoff limits



II: Pitcher evaluation from pitch tracking



III: Inference from swing-level bat tracking

Part I: Runner-pitcher strategy under pickoff limits

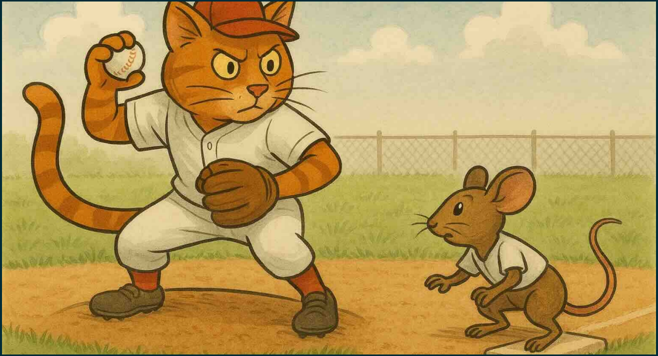
Jacob Hahn, Rice University → Chicago Cubs

Sivaramakrishnan Ramani, Rice University

Andrew Schaefer, Rice University

Working paper:
Hahn J, Powers S, Ramani S, Schaefer A "The Two-Foot Rule: A game theoretic analysis of the pickoff limit in Major League Baseball"

Background



- In 2023, MLB limited the pitcher to 3 pickoff attempts
- After 3 failed pickoffs, the runner advances automatically

The Puzzle

If you are the runner, how much should you extend your lead after each pickoff attempt?

Data: pitch-by-pitch lead distance (player tracking) from MLB*

Our Approach

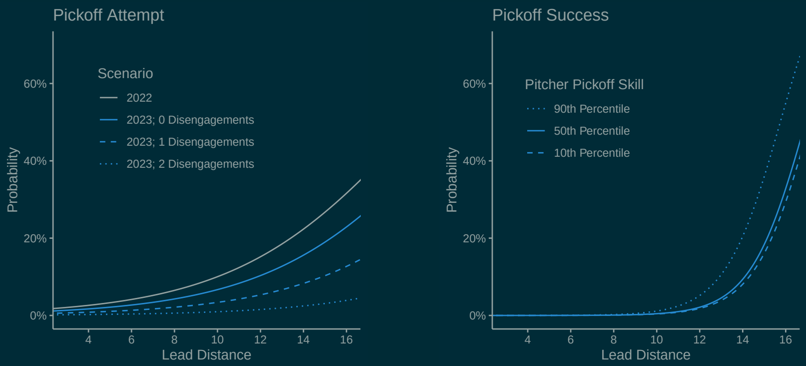
Step 1: Estimate the effect of lead distance

- Linear mixed-effects models (LMMs) for probabilities of pickoff attempt, pickoff success, stolen base success

Step 2: Solve for the optimal lead distance

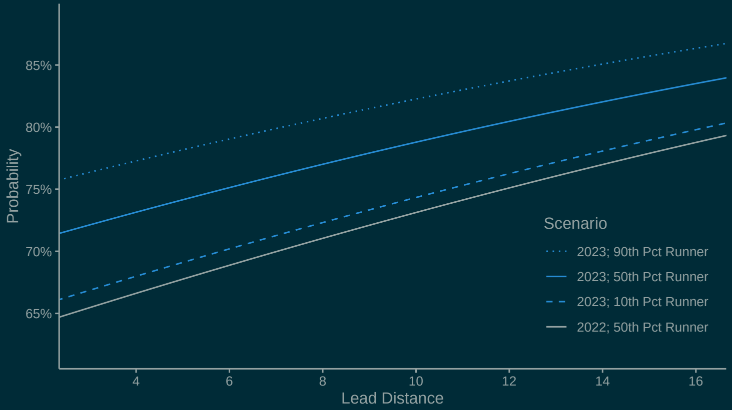
- Single-player MDP, or two-player stochastic game

Pickoff Attempt and Success Probability



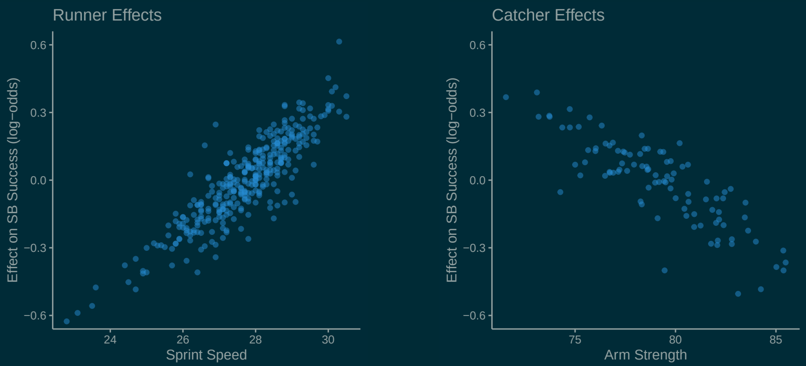
- Fixed effects: lead distance, disengagements, outs, count, runner sprint speed, catcher arm strength
- Random effects: runner, catcher, pitcher

Stolen Base Success Probability



- Fixed effects: lead distance, disengagements, outs, count, runner sprint speed, catcher arm strength
- Random effects: runner, catcher, pitcher

Runner and Catcher Effects



- Fixed effects: lead distance, disengagements, outs, count, runner sprint speed, catcher arm strength
- Random effects: runner, catcher, pitcher

Solving for Optimal Lead Distance

Two options:

1. Single-player Markov decision process (MDP)
   - Runner chooses lead distance
   - Pitcher follows known stochastic response (no agency)

2. Two-player stochastic game
   - Runner chooses lead distance
   - Pitcher chooses whether to attempt pickoff

Optimal Lead Distance (single-player MDP)

	Disengagements		
Count	1	2	3
0-0	10.0	11.5	14.1
1-0	10.2	11.7	14.2
2-0	10.2	11.7	14.2
3-0	10.3	11.8	14.2
0-1	9.8	11.4	14.1
1-1	10.0	11.6	14.2
2-1	10.3	11.8	14.3
3-1	10.4	11.9	14.3
0-2	9.5	11.1	14.1
1-2	9.9	11.4	14.3
2-2	10.0	11.5	14.2
3-2	10.4	11.9	14.4

- The Two-Foot Rule:
  Increase lead by two feet after each disengagement
- Value of implementation:
  $\sim$17 runs / team / season
  ($\sim$\$17,000,000)

Optimal Lead Distance (two-player stochastic game)

	Disengagements		
Count	1	2	3
0-0	11.2	11.8	16.2
1-0	11.2	12.0	16.1
2-0	11.0	12.2	15.9
3-0	10.4	11.6	15.3
0-1	10.8	11.3	16.2
1-1	10.9	11.1	16.1
2-1	11.0	12.1	15.9
3-1	10.5	11.6	15.7
0-2	10.2	10.8	16.2
1-2	10.5	10.4	16.3
2-2	10.4	11.4	15.9
3-2	10.3	11.1	15.6

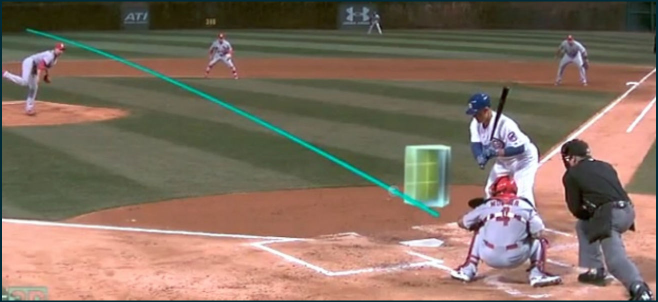
- Equilibrium entails more aggressive behavior from both runner and pitcher

Part II: Pitcher evaluation from pitch tracking

Vicente Iglesias, Susquehanna International Group

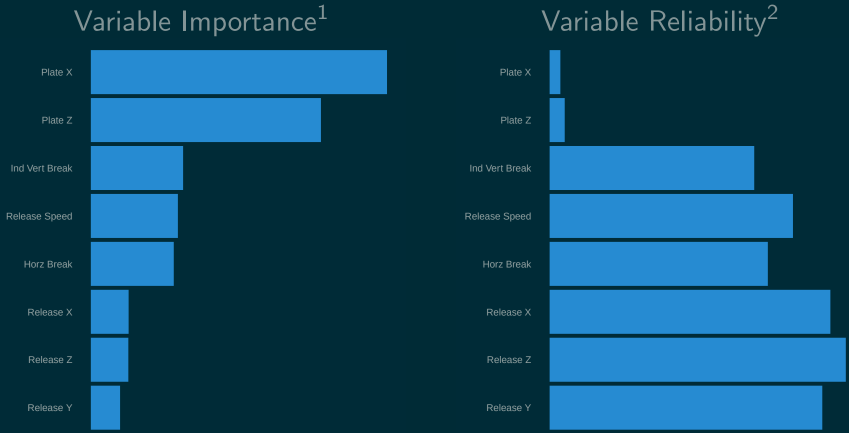
Working paper:
Powers S, Iglesias V "Pitch trajectory density estimation for predicting future outcomes"

Background



- Full pitch tracking data have been available for two decades
- Trajectory estimated as 3D quadratic function of time ($\in \mathbb{R}^9$)
- Standard practice: Evaluate pitchers based on trajectories

The Puzzle



[1] fractional contribution of each feature's splits to gradient boosting pitch model
[2] (between-pitcher variance) / (total variance); varies by pitch type (here: RHB FB)

Why Supervised Learning Isn't Enough

Supervised Learning

Pitch		Outcome
x_1	$\rightarrow$	y_1
x_2	$\rightarrow$	y_2
x_3	$\rightarrow$	y_3
	$\cdots$	
x_n	$\rightarrow$	y_n
x^*	$\rightarrow$	$\hat{y} = \hat{f}(x)$

Our Problem

Pitcher	Pitch		Outcome
A	x_1	$\rightarrow$	y_1
A	x_2	$\rightarrow$	y_2
B	x_3	$\rightarrow$	y_3
		$\cdots$	
C	x_n	$\rightarrow$	y_n
		$\searrow$	
A			$\hat{y}$

Why Supervised Learning Isn't Enough

Supervised Learning

Pitch		Outcome
x_1	$\rightarrow$	y_1
x_2	$\rightarrow$	y_2
x_3	$\rightarrow$	y_3
	$\cdots$	
x_n	$\rightarrow$	y_n
x^*	$\rightarrow$	$\hat{y} = \hat{f}(x)$

Our Approach

Pitcher	Pitch		Outcome
A	x_1	$\rightarrow$	y_1
A	x_2	$\rightarrow$	y_2
B	x_3	$\rightarrow$	y_3
		$\cdots$	
C	x_n	$\rightarrow$	y_n
	$\downarrow$		
A	$\hat{p}(x \mid A)$	$\rightarrow$	$\int \hat{p}(x)\hat{f}(x)$

Our Approach

Step 1: Estimate trajectory distributions conditional on covariates

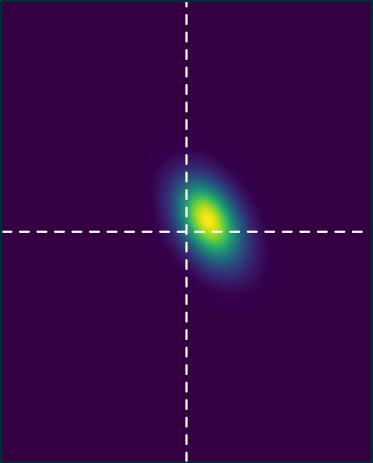
- Bayesian hierarchical model (9D MVN likelihood)

Step 2: Fit supervised learning model and integrate w.r.t. Step 1

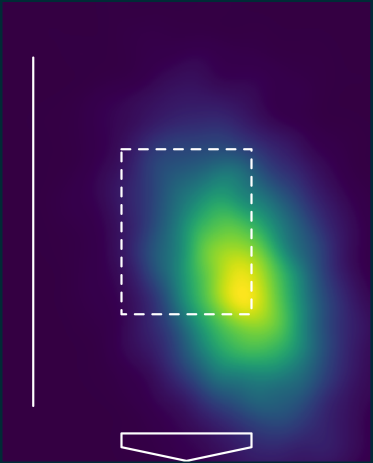
- Gradient boosting

Dylan Cease's Slider vs RHB in 0-0 Counts

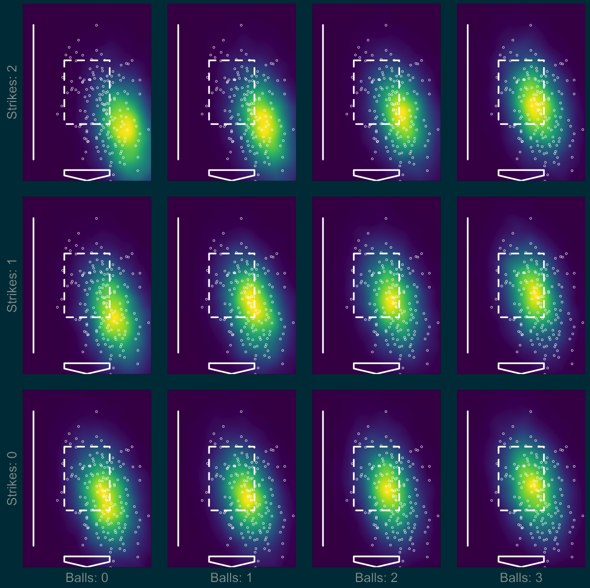
Predicted Break Chart



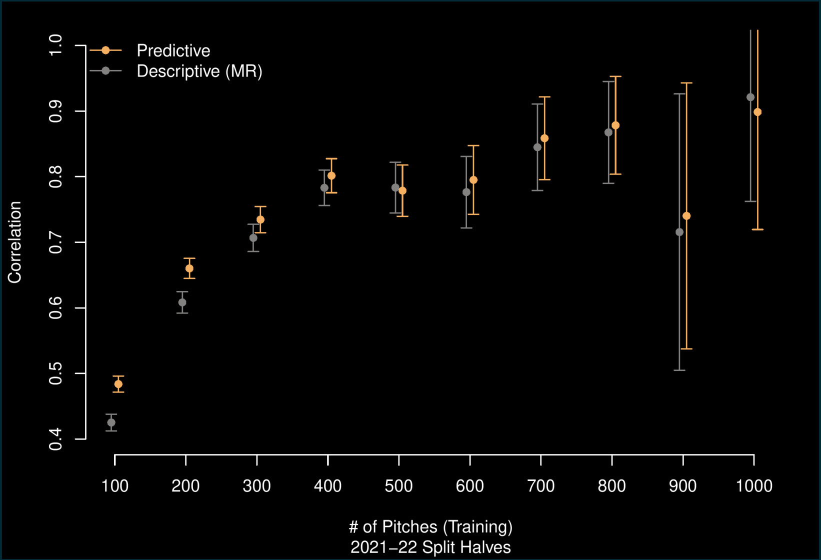
Predicted Plate Location



Dylan Cease's Slider vs RHB in All Counts



Predictive Performance



Part III: Inference from swing-level bat tracking

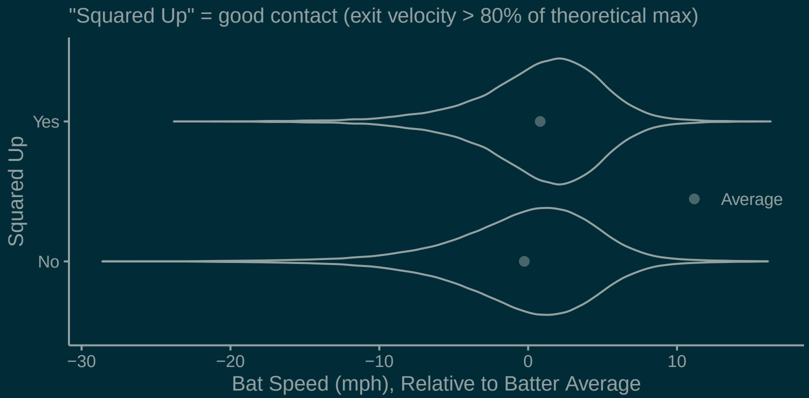
Ron Yurko, Carnegie Mellon University

Working paper:
Powers S, Yurko R “Swinging, Fast and Slow: Interpreting variation in baseball swing tracking metrics” arXiv:2507.01238

Background

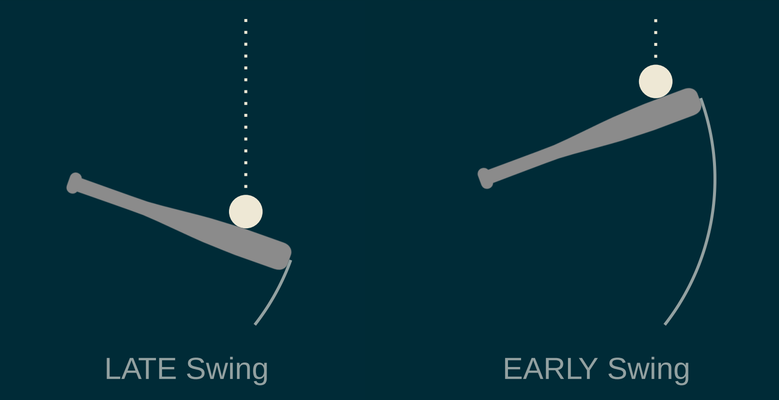
- In May 2024, MLB released swing length and bat speed data
- Both are measured at contact (or at point nearest to contact)

The Puzzle



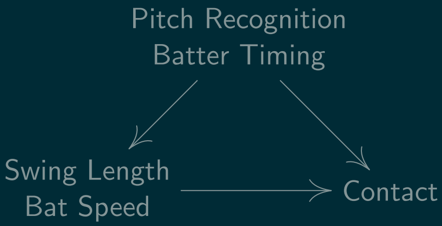
- How should batters modulate swing length and bat speed?

Challenge #1: Measurement



- For identical swings, timing determines point of measurement
- Swing length and bat speed are outcomes, not purely process

Challenge #2: Confounding



Our Approach

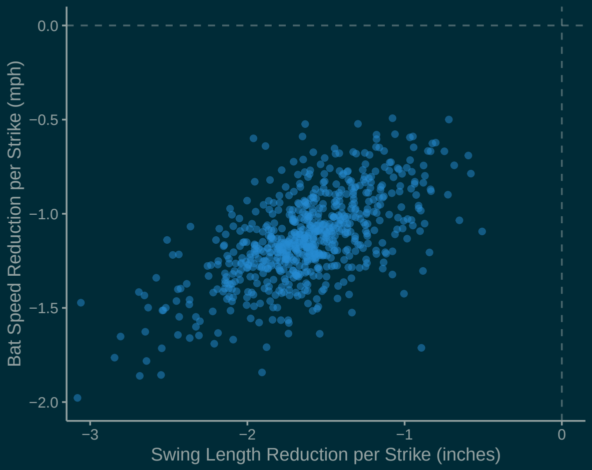
Step 1: Estimate batter intention

- Bayesian hierarchical model (skew-normal likelihood)
- Condition on successful contact against fastballs!

Step 2: Estimate effect of batter intention

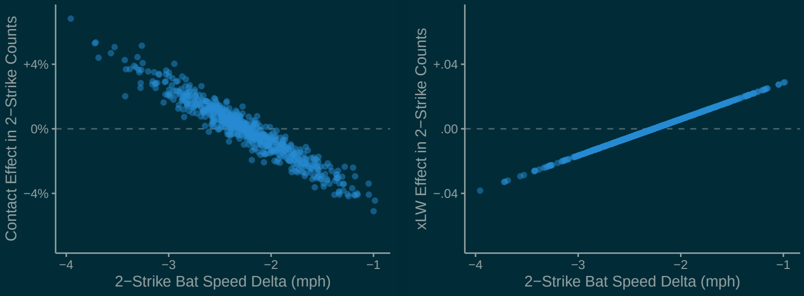
- Instrumental variable regression (instrument: count)

Batter × Count Interaction



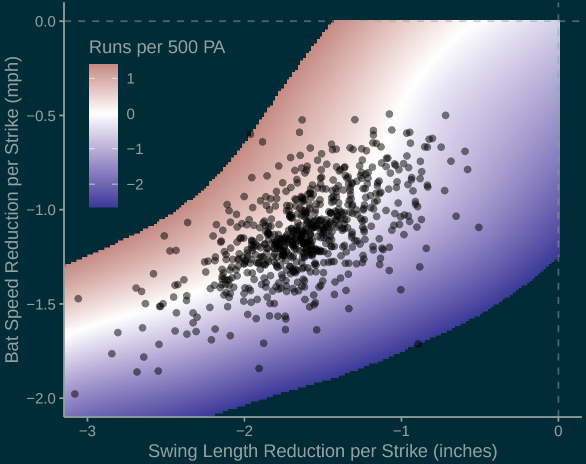
- Effects: batter, count, location, batter × (count, location)
- We observe heterogeneity in how batters adjust to count

Causal Effect Estimates



- Batters who slow down more with two strikes exhibit more contact (relative to expectation) with two strikes
- Batters who slow down more with two strikes exhibit less power (relative to expectation) with two strikes
- Is the tradeoff worthwhile?

Contact-Power Tradeoff



- Batters can reduce strikeouts by slowing their bat speeds, but the tradeoff in power makes it a wash

Thank You!

saberpowers.github.io